

GML_04_02-equities.R

frank

2024-05-06

```
#feb 2023, FL

# STEP 1

# Clean previous variables

rm(list = ls())

library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

#library(egcm)

#diese auskommentierte Sektion zeigt, wo die Daten herkommen
#beginDate = as.Date("2010-01-01")
#endDate = as.Date("2016-01-01")
#quote = c("Open", "High", "Low", "Close","Volume")
#compression = "d"
#provider = "yahoo"
#retclass = c("zoo", "its", "ts")
# Daten runterladen
#thyssenkrupp = get.hist.quote(instrument='TKA.DE',start = beginDate, end =
endDate,
  #                               quote = quote, provider = provider, method =
NULL,
  #                               origin = "1899-12-30", compression =
compression,
  #                               retclass = retclass, quiet = TRUE, drop =
FALSE)
#thyssenkruppStock = thyssenkrupp

#SalzgitterAG = get.hist.quote(instrument='SZG.DE',start = beginDate, end =
endDate,
  #                               quote = quote, provider = provider, method =
NULL,
  #                               origin = "1899-12-30", compression =
compression,
  #                               retclass = retclass, quiet = TRUE, drop =
FALSE)
#SalzgitterStock = SalzgitterAG
```

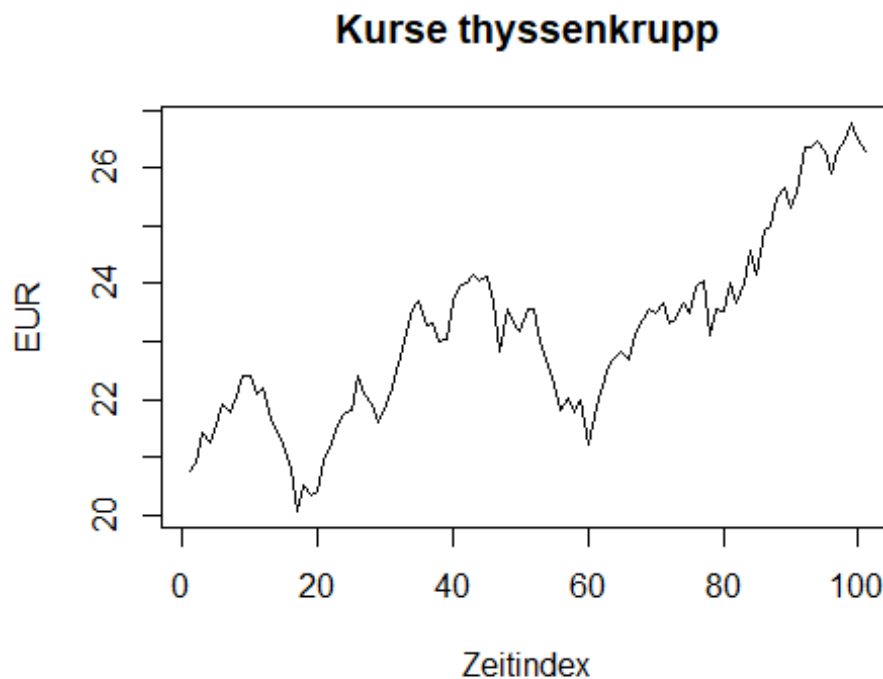
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#DAX = get.hist.quote(instrument='^GDAXI',start = beginDate, end = endDate,
#                      quote = quote, provider = provider, method =
NULL,
#                      origin = "1899-12-30", compression =
compression,
#                      retclass = retclass, quiet = TRUE, drop =
FALSE)
#DAXIndex = DAX
#Mergen
#zDataPre <- merge(thyssenkruppStock, SalzgitterStock, DAXIndex)
#zDataPrice <- zDataPre[complete.cases(zDataPre)]
#nrow(zDataPrice)
#store equity data for following exercises
#OUTPUT<-zDataPrice
#write.csv2(OUTPUT, file="xxx.csv") #write.csv2 uses a comma for the
decimal point

data_table <- read.table("GML_04_02 equities.csv", header=TRUE,
sep=";",dec=",")
TK <- data_table[,2][111:211]
SG <- data_table[,7][111:211]

x11()
plot(TK, ylab = "EUR", xlab = "Zeitindex", type='l', main = "Kurse
thyssenkrupp")

```

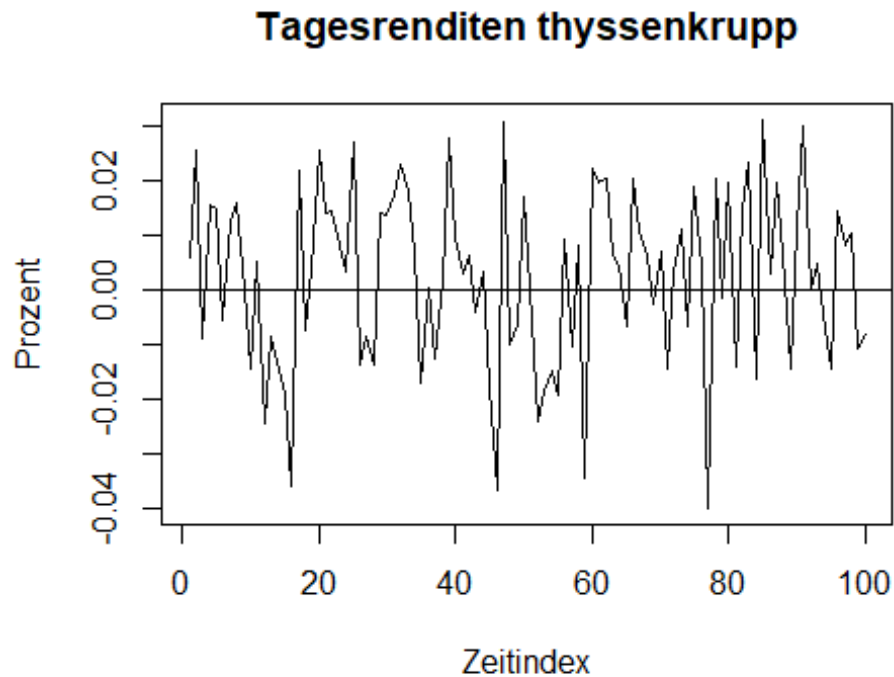


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x11()
plot(diff(log(TK)),ylab = "Prozent", xlab = "Zeitindex",type='l', main =

```

```
"Tagesrenditen thyssenkrupp")  
abline(h=0)
```



```
# STEP 2  
length(TK)  
## [1] 101  
TS = TK  
  
# Apply usual tests  
adf.test(TS)#H0 Non-stat  
  
##  
## Augmented Dickey-Fuller Test  
##  
## data: TS  
## Dickey-Fuller = -2.1513, Lag order = 4, p-value = 0.514  
## alternative hypothesis: stationary  
  
pp.test(TS)  
  
##  
## Phillips-Perron Unit Root Test  
##  
## data: TS  
## Dickey-Fuller Z(alpha) = -9.3001, Truncation lag parameter = 4, p-value
```

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## = 0.5767
## alternative hypothesis: stationary

kpss.test(TS)#H0 Stat

## Warning in kpss.test(TS): p-value smaller than printed p-value

##
## KPSS Test for Level Stationarity
##
## data: TS
## KPSS Level = 1.475, Truncation lag parameter = 4, p-value = 0.01

# STEP 3

#make sure same degree of integration = I(1)
# Apply usual tests TK:
TS = diff(TK)
adf.test(TS)#H0 Non-stat

## Warning in adf.test(TS): p-value smaller than printed p-value

##
## Augmented Dickey-Fuller Test
##
## data: TS
## Dickey-Fuller = -4.4152, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary

pp.test(TS)

## Warning in pp.test(TS): p-value smaller than printed p-value

##
## Phillips-Perron Unit Root Test
##
## data: TS
## Dickey-Fuller Z(alpha) = -106.84, Truncation lag parameter = 3, p-value
## = 0.01
## alternative hypothesis: stationary

kpss.test(TS)#H0 Stat

## Warning in kpss.test(TS): p-value greater than printed p-value

##
## KPSS Test for Level Stationarity
##
## data: TS
## KPSS Level = 0.074314, Truncation lag parameter = 4, p-value = 0.1

# Apply usual tests SG:
TS = diff(SG)
adf.test(TS)#H0 Non-stat

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## Warning in adf.test(TS): p-value smaller than printed p-value

##
## Augmented Dickey-Fuller Test
##
## data: TS
## Dickey-Fuller = -4.4483, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary

pp.test(TS)

## Warning in pp.test(TS): p-value smaller than printed p-value

##
## Phillips-Perron Unit Root Test
##
## data: TS
## Dickey-Fuller Z(alpha) = -112.56, Truncation lag parameter = 3, p-value
## = 0.01
## alternative hypothesis: stationary

kpss.test(TS)#H0 Stat

## Warning in kpss.test(TS): p-value greater than printed p-value

##
## KPSS Test for Level Stationarity
##
## data: TS
## KPSS Level = 0.050905, Truncation lag parameter = 4, p-value = 0.1

#confirmed

# STEP 4

#check a la Engle Granger
reg = lm(SG ~ TK)
summary(reg)

##
## Call:
## lm(formula = SG ~ TK)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.2570 -1.0381  0.0349  0.9835  4.6196
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   32.9719     2.8195  11.694 < 2e-16 ***
## TK              0.7670     0.1216   6.309 7.94e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

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## Residual standard error: 1.959 on 99 degrees of freedom
## Multiple R-squared:  0.2867, Adjusted R-squared:  0.2795
## F-statistic: 39.8 on 1 and 99 DF, p-value: 7.938e-09

resis = residuals(reg)#if cointegrated should have nice R²!
# Apply ordinary tests
# Apply usual tests SG:
TS = resis
adf.test(TS)#H0 Non-stat

##
## Augmented Dickey-Fuller Test
##
## data: TS
## Dickey-Fuller = -3.3387, Lag order = 4, p-value = 0.06854
## alternative hypothesis: stationary

pp.test(TS)

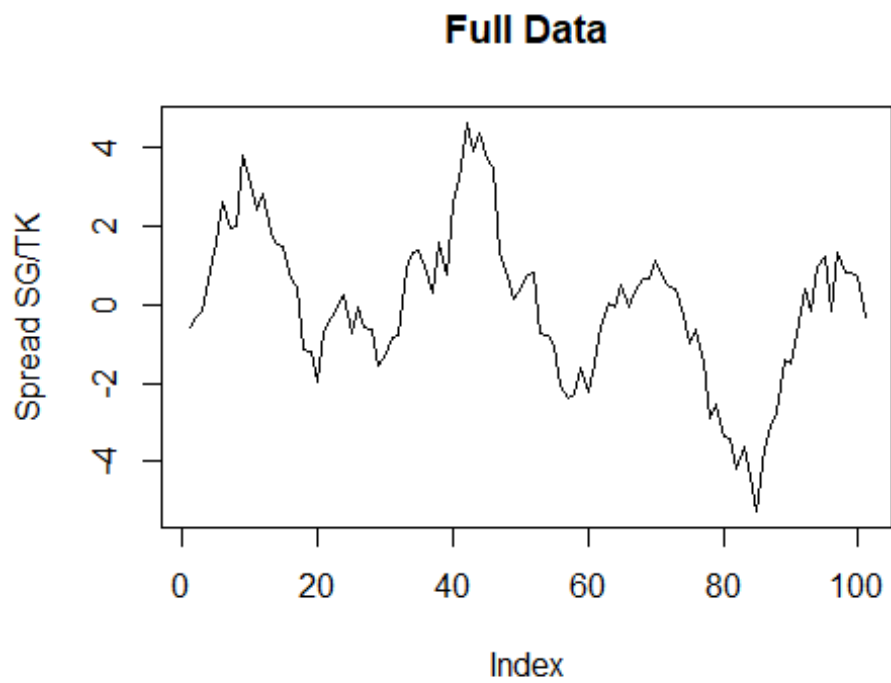
##
## Phillips-Perron Unit Root Test
##
## data: TS
## Dickey-Fuller Z(alpha) = -14.602, Truncation lag parameter = 4, p-value
## = 0.2685
## alternative hypothesis: stationary

kpss.test(TS)#H0 Stat

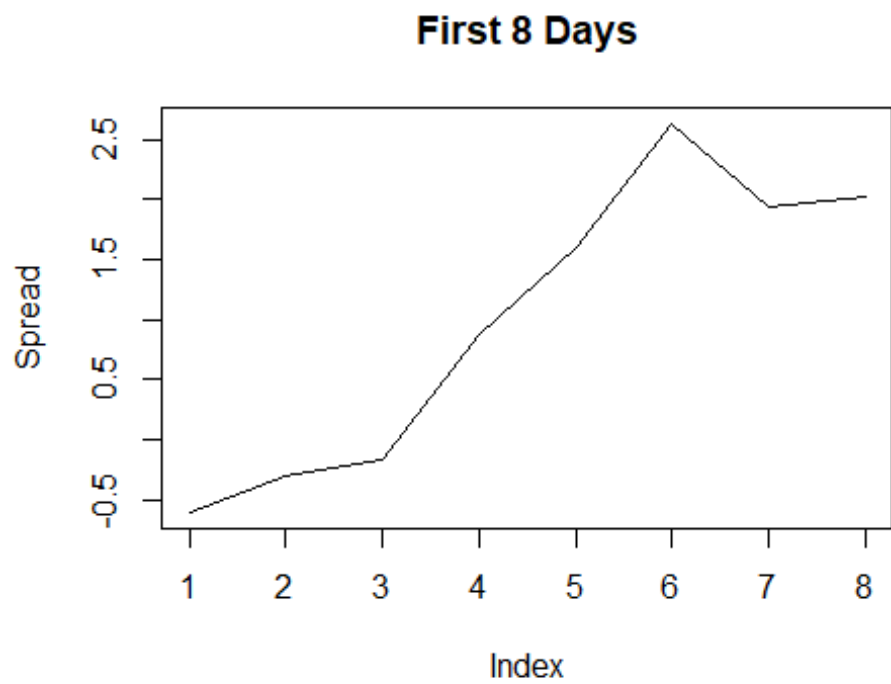
##
## KPSS Test for Level Stationarity
##
## data: TS
## KPSS Level = 0.49953, Truncation lag parameter = 4, p-value = 0.04177

#plotting residuals
x11()
plot(residuals(reg), type='l', ylab = "Spread SG/TK", main = "Full Data")

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x11()  
plot(residuals(reg)[1:8], type='l', ylab = "Spread", main = "First 8 Days")
```



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# STEP 5
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#go for ECM

#take resi t-1
lagged_resis = residuals(reg)[-length(residuals(reg))]

regECM = lm(diff(SG)~diff(TK)+lagged_resis-1)
summary(regECM)#as hoped for the coef of ECterm is negative

##
## Call:
## lm(formula = diff(SG) ~ diff(TK) + lagged_resis - 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.10077 -0.52319 -0.07532  0.36043  1.60884
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## diff(TK)         1.96378     0.18329  10.714  <2e-16 ***
## lagged_resis -0.03011     0.03491  -0.862    0.391
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6604 on 98 degrees of freedom
## Multiple R-squared:  0.565, Adjusted R-squared:  0.5561
## F-statistic: 63.63 on 2 and 98 DF, p-value: < 2.2e-16

head(cbind(TK,SG))

##           TK      SG
## [1,] 20.775 48.305
## [2,] 20.900 48.700
## [3,] 21.440 49.250
## [4,] 21.245 50.150
## [5,] 21.580 51.120
## [6,] 21.900 52.400

# Apply ordinary tests
adf.test(resis )#H0 Non-stat

##
## Augmented Dickey-Fuller Test
##
## data:  resis
## Dickey-Fuller = -3.3387, Lag order = 4, p-value = 0.06854
## alternative hypothesis: stationary

pp.test(resis )

##
## Phillips-Perron Unit Root Test
##

```

```

## data:  resis
## Dickey-Fuller Z(alpha) = -14.602, Truncation lag parameter = 4, p-value
## = 0.2685
## alternative hypothesis: stationary

kpss.test(resis )#H0 Stat

##
## KPSS Test for Level Stationarity
##
## data:  resis
## KPSS Level = 0.49953, Truncation lag parameter = 4, p-value = 0.04177

#nevertheless assume coint, based on ADF an pedagogics

#noch ein Kointegrationstest: H0 not cointegrated.
po.test(as.matrix(cbind(TK,SG)))

## Warning in po.test(as.matrix(cbind(TK, SG))): p-value greater than printed
## p-value

##
## Phillips-Ouliaris Cointegration Test
##
## data:  as.matrix(cbind(TK, SG))
## Phillips-Ouliaris demeaned = -0.5108, Truncation lag parameter = 1,
## p-value = 0.15

#a final word
#egcm(SG,TK)#use this also in live

#Ende

```